Lecture 4: The language of probability

- This lecture will cover foundational material, mainly definitions, for statistical modelling.
- Amont the concepts encountered, you'll hear about outcomes, events, probability, random variables, distributions.
- I will assume you know about mean, variance, correlation, etc. Not just the empirical versions computed from the data but the parameters associated to distributions as well.

Probability Models

A probability model is a mathematical device for describing random phenomena. It has two parts:

- A set of possible **outcomes** called the **sample space** (usually denoted S).
- A **probability** for each outcome (or density for ranges in the continuous case) measuring the **plausibility** of the outcome.

Example 1: Coin tossing
Toss a coin. The sample space is $S = \{\text{heads}, \text{tails}\}$, or more briefly, $S = \{H, T\}$.

Sample Spaces

We have freedom to choose a sample space that is convenient for the particular problem at hand.

Example 2: More coin tossing
Toss a coin 3 times and record the results in order. The sample space is

$$S = \{TTT, TTH, THT, HTT, THH, HTH, HHT, HHH\}$$

using the obvious notation.

If we were only interested in the number of heads, then a suitable (simpler) sample space would be $S = \{0, 1, 2, 3\}$.

Sample Spaces

continued

The previous Examples involves *finite* sample spaces. Not all samples spaces are finite.

Example 3: Unlimited coin tossing
Toss a coin repeatedly until first head is observed. The sample space is

$$S = \{H, TH, TTH, TTTH, TTTTH, \ldots\}.$$

Example 4: Rainfall measurements
Record the exact rainfall at Palmerston North airport next month. A suitable sample space is $S = \{\text{any non-negative number}\}$.

Events

An **event** is a subset of the sample space (i.e. an outcome or set of outcomes).

Example 5: Two heads are better than one?!
Let A be the event that exactly 2 heads are observed. Then $A = \{HHT, HTH, THH\}$.

Example 6: Rainfall event
For the rainfall Example, $A = \{\text{rainfall} \geq 300\text{mm}\}$ is an event.

Probability Rules

For any event A, probability $\mathrm{P}(A)$ of A must satisfy:

(i) $0 \leq \mathrm{P}(A) \leq 1$.

(ii) $\mathrm{P}(S) = 1$ for sample space S.

(iii) **Complement** of event A is the event that A does not occur, written A^c. Must have

$$\mathrm{P}(A^c) = 1 - \mathrm{P}(A)$$

(iv) Events A and B are **disjoint** (or **mutually exclusive**) if they have no outcomes in common (cannot occur together). For disjoint A, B,

$$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B)$$

where $A \cup B$ is the event 'A or B occurs'.

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Probability Rules

continued

If events A and B are not disjoint then final probability rule needs revising.

(iv)' For events A and B,

$$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$$

where $A \cap B$ is the event 'both A and B occur'.

Do a Venn diagram!

Independence

- Informally, events A and B are independent if occurrence of one tells you nothing about whether the other has occurred.
- In mathematical terms, events A and B are independent when

$$\mathrm{P}(A \cap B) = \mathrm{P}(A)\mathrm{P}(B)$$

- Extends in a simple fashion to three or more independent events.

Example 7: Triple coin toss
In three independent tosses of a fair coin, the probability of HTH (in that specific order) is

$$\mathrm{P}(HTH) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$$

Conditional Probability

- Often natural to think of probability of an event conditional on occurrence of another event.
- We write $\mathrm{P}(A|B)$ for the **conditional probability** of event A given event B has occurred.

Example 8: Aces

Two cards are drawn, without replacement, from a deck.

Let $B = \{\text{first draw is an ace}\}$ and $A = \{\text{second draw is an ace}\}$.

Then $\mathrm{P}(A|B) = 3/51$, which is different to $\mathrm{P}(A) = 4/52 = 1/13$.

Properties of Conditional Probability

(i) For events A and B,

$$\mathrm{P}(A \cap B) = \mathrm{P}(A|B)\mathrm{P}(B),$$

where $\mathrm{P}(A \cap B)$ is the probability of both A and B occurring.

(ii) From (i)

$$\mathrm{P}(A|B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$$

so long as $\mathrm{P}(B) > 0$.

(iii) If A and B are independent then $\mathrm{P}(A \cap B) = \mathrm{P}(A)\mathrm{P}(B)$ and so $\mathrm{P}(A|B) = \mathrm{P}(A)$. If $\mathrm{P}(A|B) \neq \mathrm{P}(A)$, then A and B are dependent.

Example 9: Aces again

In the previous Example we had $\mathrm{P}(B) = 4/52 = 1/13$. Hence, the probability of getting aces with both draws is given by

$$\mathrm{P}(A \cap B) = \mathrm{P}(A|B)\mathrm{P}(B) = \frac{3}{51} \times \frac{1}{13} = \frac{1}{221}.$$

Law of Total Probability

Simple version of Law of Total Probability

$$\mathrm{P}(A) = P(A|B)\mathrm{P}(B) + P(A|B^c)P(B^c).$$

Generalized version

Let $B_1, B_2, \ldots, B_n$ be a **partition** of the sample space S. This means that:

- $B_1, B_2, \ldots, B_n$ are **mutually exclusive**: i.e. $B_i \cap B_j = \emptyset$; and
- $B_1, B_2, \ldots, B_n$ are **exhaustive**: i.e. $B_1 \cup B_2 \cup \ldots \cup B_n = S$.

Then

$$\mathrm{P}(A) = \sum_{i=1}^{n} \mathrm{P}(A|B_i)\mathrm{P}(B_i).$$

Do a Venn diagram again to interpret conditional probabilities!

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Example 10: Deadly Drooling Disease

Question

90% of people with Deadly Drooling Disease have some unusual symptoms. The probability of someone without this disease showing these symptoms is 0.001. If 1 person in a million has DDD, what is the probability that a randomly selected person exhibits the symptoms?

Solution

Let $A = \{\text{Has Deadly Drooling Disease}\}$ and $B = \{\text{Has symptoms}\}$.

Then $\mathrm{P}(A) = 0.000001$, $\mathrm{P}(B|A) = 0.9$ and $\mathrm{P}(B|A^c) = 0.001$.

Law of Total Probability (with A, B interchanged) gives

$$\begin{aligned} \mathrm{P}(B) &= \mathrm{P}(B|A)\mathrm{P}(A) + \mathrm{P}(B|A^c)\mathrm{P}(A^c) \\ &= 0.9 \times 0.000001 + 0.001 \times 0.999999 \\ &= 0.001000899 \end{aligned}$$

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Bayes Theorem

Bayes' Theorem
Let A and B be events. Then

$$\mathrm{P}(A|B) = \frac{\mathrm{P}(B|A)\mathrm{P}(A)}{\mathrm{P}(B)}.$$

An extended version of the result is often useful.

Bayes' Theorem (extended version)
Let $A_1, A_2, \ldots, A_n$ be a partition of sample space S, and B be some other event. Then

$$\mathrm{P}(A_i|B) = \frac{\mathrm{P}(B|A_i)\mathrm{P}(A_i)}{\sum_{j=1}^{n} \mathrm{P}(B|A_j)\mathrm{P}(A_j)}.$$

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Example 11: Hypochondria and Bayes' Theorem

Question:
A hypochondriac finds that he has the usual symptoms for Deadly Drooling Disease. Should he be worried?

Solution:
Let $A = \{\text{Has Deadly Drooling Disease}\}$ and $B = \{\text{Has symptoms}\}$.

Then $\mathrm{P}(A) = 0.000001$ (prior); $\mathrm{P}(B|A) = 0.9$; $\mathrm{P}(B|A^c) = 0.001$.

Also, $\mathrm{P}(A^c) = 1 - \mathrm{P}(A) = 0.999999$ and $\mathrm{P}(B) = 0.001000899$ from previous Example.

Hence

$$\mathrm{P}(A|B) = \frac{\mathrm{P}(B|A)\mathrm{P}(A)}{\mathrm{P}(B)} = \frac{0.9 \times 0.000001}{0.001000899}$$
$$= 0.000899 \text{ (3 sig. fig.)}$$

What is a Random Variable?

- The outcome in a probability model is drawn from the sample space, S.
- This can include a variety of types of outcomes:
  - E.g. $S = \{H, T\}$ for coin tossing experiment;
  - E.g. $S = [0, \infty)$ rainfall tomorrow in Palmerston North.
- To allow mathematical analysis, convenient to look at numerical outcomes.
- A **random variable** assigns a numerical value to each possible outcome (i.e. each element of S).
  - E.g. For $S = \{H, T\}$, assign 1 for H and 0 for T
  - E.g. For $S = [0, \infty)$, outcome is numeric, so random variable can equal outcome itself.
- This is the **data before you see it**.

Notation for Random Variables

- Random variables usually denoted by upper case letters towards the end of the alphabet.
  - E.g. X, Y, Z (any one will do).
  - E.g. For $\{H, T\}$, $X = 1$ if outcome is H and $X = 0$ if T.
- Possible fixed values taken by random variable are denoted by lower case letters.
  - E.g. x, y, z (any one will do).
  - Notice the distinction!
- Later, can think of x as an observed piece of data.
- Corresponding random variable X can be thought of as 'data waiting to happen' (still uncertain, and hence governed by probabilities).
- x is a 'realized' value (or 'realization') of X.

Discrete Random Variables

Discrete Random Variable

A random variable (r.v.) X is discrete if it can take either:

- a finite number of values; or
- an infinite number of whole number values (or equivalent).

Example 12: Coin tossing

Single coin tossed. $X = 1$ if head, $X = 0$ if tail. Then X is a discrete random variable.

Example 13: Another coin tossing random variable

Let X be the number of tails obtained in tosses of a coin until first head is obtained. Then X is a discrete random variables, taking possible values $0, 1, 2, 3, \ldots$.

Continuous Random Variables

Continuous Random Variable
A continuous random variable X can take all values in an interval of numbers.

Example 14: Random rainfall
Define random variable X to equal rainfall (in mm) to fall at Palmerston North airport next month.

Example 15: Earthquake!
Define random variable Y to to be time until next earthquake of magnitude 4.0 or greater to strike NZ.

Probability Mass Function

For Discrete Random Variables

- Distribution of a discrete random variable X is described by its **probability (mass) function**, f.
- This gives the probabilities for all possible values that X can take.
  - Will write $x_1, x_2, \ldots, x_k$ for these values.

Value of X	x_1	x_2	$\ldots$	x_k
Probability	$f(x_1)$	$f(x_2)$	$\ldots$	$f(x_k)$

- Points to note:
  - $f(x_i) > 0$ for $i = 1, 2, \ldots, k$;
  - $\sum_i f(x_i) = 1$.

Example 16: Double coin toss

- A fair coin is tossed twice.
- Let X be the number of heads recorded.
  - Hence possible outcomes are 0, 1, 2.
- Using independence of results of the two tosses, we calculate

$$
\begin{aligned}
f(0) &= \mathrm{P}(X=0) = \mathrm{P}(TT) = \frac{1}{2}\times\frac{1}{2} = \frac{1}{4} \\
f(1) &= \mathrm{P}(X=1) = \mathrm{P}(TH) + \mathrm{P}(HT) = \frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2} = \frac{1}{2} \\
f(2) &= P(X=2) = \mathrm{P}(HH) = \frac{1}{2}\times\frac{1}{2} = \frac{1}{4}
\end{aligned}
$$

- Hence probability mass function is

x	0	1	2
$f(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

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Example 17: Waiting Game

- A biased coin has probability p of landing heads.
- Let X be the number of tails obtained before the first head.
  - X can take any non-negative integer value: $0, 1, 2, \ldots$
- $$\begin{aligned} \mathrm{P}(X = 0) &= \mathrm{P}(H) = p \\ \mathrm{P}(X = 1) &= \mathrm{P}(TH) = (1-p)p \\ &\vdots \quad\quad \vdots \\ \mathrm{P}(X = k) &= \mathrm{P}(T \ldots TH) = (1-p)^k p \\ &\vdots \quad\quad \vdots \end{aligned}$$
- Hence the probability mass function of X is given by

x	0	1	$\ldots$	k	$\ldots$
$f(x)$	p	$(1-p)p$	$\ldots$	$(1-p)^k p$	$\ldots$

Probability Density Functions

For continuous random variables

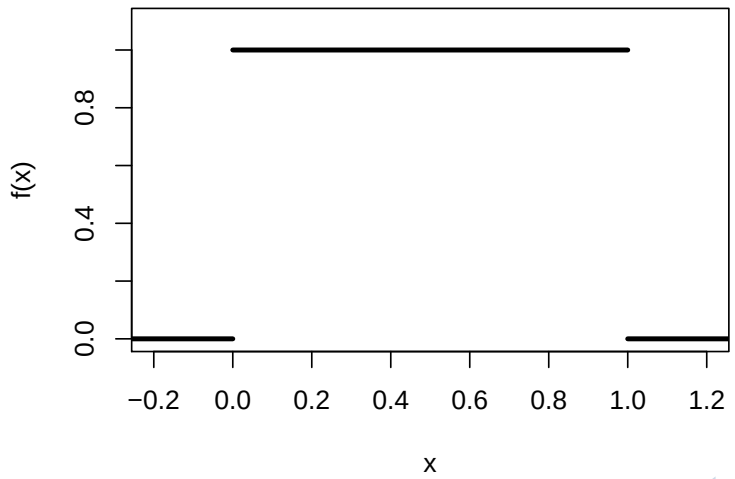
- The distribution of a continuous random variable is described by a **probability density (function)**, f.
- This is designed to provide probabilities for intervals of values, rather than individual outcomes.
  - Sensible. Interesting to know whether rainfall will be e.g. greater than 100mm, or 60 to nearest mm.
- A probability density function f for random variable X has the following properties:
  - $f(x) \geq 0$ for all x; that is, a density function never takes negative values.
  - $\mathrm{P}(a \leq X \leq b) = \int_a^b f(x)\,dx = \mathrm{P}(a < X < b)$.
  - I.e. the area under the curve between a and b gives $\mathrm{P}(a \leq X \leq b)$.
  - Follows that $\int_{-\infty}^{\infty} f(x)\,dx = 1$: total area under a density curve is one.

Example 18: Uniformly Distributed

- Continuous random variable X has a **uniform distribution** on the interval $[0, 1]$ if its probability density is given by

$$f(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

- Probability density function displayed below.
- Write $X \sim \text{Unif}(0, 1)$ as a shorthand notation to indicate that X follows this distribution.
- More generally, $X \sim \text{Unif}(a, b)$ if the random variable X follows a uniform distribution on the interval (a, b).



Example 18 continued

- Suppose we want to compute $P(0.3 < X < 0.8)$.
- This probability given by area under the density curve below.
- Can be computed by basic geometry.
  - $P(0.3 < X < 0.8) =$ area of rectangle $= (0.8 - 0.3) \times 1 = 0.5$
- Can also be computed using integration:

$$P(0.3 < X < 0.8) = \int_{0.3}^{0.8} f(x)\,dx = \int_{0.3}^{0.8} 1\,dx = [x]_{0.3}^{0.8} = 0.5$$

- Will leave such calculations to the computer in 161.304.

